

Day 33 - IPV6 (pt 3)

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IPv6 Information

IPv6 Header

The IPv6 header is much simpler than the IPv4 header.

- Two of the biggest differences are
 - The amount of fields in the header, as it has less fields than the **IPv4** header. This helps with performance.
 - The fixed header size of **40 bytes** for **IPv6**.

IPv6 Header - Fields

The IPv6 header consists of 8 fields in total, being:

1. **Version (4 bits)**
 - Indicates the version of IP that is used.
 - Fixed value of **6 (0b0110)** to indicate **IPv6**
2. **Traffic Class (8 bits)**
 - Used for **QoS (Quality of Service)** to indicate high-priority traffic
3. **Flow Label (20 bits)**
 - Used to identify specific traffic 'flows' (communications between a specific source and destination)
4. **Payload Length (16 bits)**
 - Indicates the length of the payload (the encapsulated Layer 4 **segment**) in bytes.
5. **Next Header (8 bits)**
 - Indicates the type of the 'next header' (header of the encapsulated **segment**), for example **TCP** or **UDP**.
 - Same function as the **IPv4** header's '**protocol**' field
6. **Hop Limit (8 bits)**
 - Value is decremented by **1** for each router that forwards it. If it reaches **0**, the packet is discarded.
7. **Source Address (128 bits)**
8. **Destination Address (128 bits)**

Solicited-Node Multicast Address

An IPv6 **solicited-node multicast** address is calculated from a unicast address through the following calculation:

- **ff02::1:ff + the last 6 hex digits from a unicast address**
Example of the solicited-node multicast address calculation

- **Example 1:**

- **Unicast address:** 2001 : 0db8 : 0000 : 0001 : 0f2a : 4fff : fea3 : 00b1
- We take the last 6 **hex** digits and add them onto **ff02::1:ff** and get:
- **ff02 :: 1 : ffa3 : 00b1**

- **Example 2:**

- **Unicast address:** 2001 : 0db8 : 0000 : 0001 : 0489 : 4eda : 073a : 12b8
- We take the last 6 **hex** digits and add them onto **ff02::1:ff** and get:
- **ff02 :: 1: ff3a : 12b8**

Neighbor Discovery Protocol (NDP)

Neighbor Discovery Protocol (NDP) is a protocol used with IPv6. It was made to replace ARP which is no longer used in IPv6.

- It uses **ICMPv6** and **solicited-node multicast** addresses to learn the MAC address of other hosts.
- Two message types are used:
 1. **Neighbor Solicitation (NS)** = **ICMPv6** Type **135**
 2. **Neighbor Advertisement (NA)** = **ICMPv6** Type **136**

- Another function of **NDP** is that it allows hosts to automatically discover routers on the local network.
 1. **Router Solicitation (NS)** = **ICMPv6** Type **133**
 2. **Router Advertisement (NA)** = **ICMPv6** Type **134**

IPv6 Neighbor Table

IPv6 doesn't have an ARP Table as it doesn't use ARP. However, it does have a neighbor table that is accessible through the command `show ipv6 neighbor`

Neighbor Solicitation (NS)

When an NS is sent, it uses a few different addresses:

- The **source IP** is set as the sending interface's IP
- The **destination IP** is the destination's **solicited-node multicast** address.
- The **source MAC** is the sending interface's MAC
- The **destination MAC** is the destination's **multicast MAC** address, based on the destination's **solicited-node multicast** address.

Neighbor Advertisement (NA)

When an NA is sent, it uses normal address values:

- The **source IP** is set as the sending interface's IP
- The **destination IP** is the destination's IP address
- The **source MAC** is the sending interface's MAC
- The **destination MAC** is the destination's MAC address

Router Solicitation (RS)

Sent when an interface is enabled or when a host connects to a network.

- Sent to **multicast** address **FF02::2** (**all routers**).
- Asks all routers on the local link to identify themselves.

Router Advertisement (RA)

These messages are sent in response to RS messages, and sometimes periodically.

- Sent to **multicast** address **FF02::1** (**all nodes**).
- The router announces its presence, as well as other information about the network.
- They are also sent periodically, even if the router hasn't received an **RS**.

Stateless Address Auto-Configuration (SLAAC)

Stateless Address Auto-Configuration (SLAAC) is a new functionality of IPv6 that allows hosts to use the RS/RA messages to learn the IPv6 prefix of the local link, and then automatically generate an IPv6 address.

- When you use the `ipv6 address PREFIX/PREFIX-LENGTH eui-64` command, you need to manually specify the prefix.
- However, using the `ipv6 address autoconfig` command, the device uses **NDP** to learn the prefix on the local link, and will automatically generate an **IPv6** address. (The device will use **EUI-64** to generate the interface identifier, or it will be randomly generated.)

Duplicate Address Detection (DAD)

Duplicate Address Detection (DAD) allows hosts to check if other devices on the local link are using the same IPv6 address

- **DAD** uses two messages to perform this check:
 1. **NS**
 2. **NA**
- Anytime an **IPv6**-enabled interface initializes, or an **IPv6** address is configured on an interface through any method, it performs **DAD**
- The host will send an **NS** to its own **solicited-node multicast IPv6** address. If it doesn't get a reply, it knows the address is **unique**.
- However, if it gets a reply, it means the address is already **in use** by another host on the network.

IPv6 Routing

The way that IPv6 routing functions is the same as IPv4.

However, the two processes are separate on the router and have their own separate routing tables as well

however, the two processes are separate on the router and have their own separate routing tables as well.

- **IPv4** routing is enabled by default
- **IPv6** routing is disabled by default, and must be enabled through `ipv6 unicast-routing`
 - If **IPv6** routing is disabled, that means the router will be able to send and receive **IPv6** traffic, but will not be able to route/forward **IPv6** traffic.

Definitions reminder:

- **Connected network route**: A route that is automatically added for each connected network.
- **Local host route**: A route that is automatically added for each address configured on the router.

IPv6 route tables and Link-local addresses

Routes for link-local addresses are not added to the routing table.

IPv6 Static Routing

In IPv6 *static routes* are setup through this command:

`ipv6 route DESTINATION/PREFIX-LENGTH {NEXT-HOP | EXIT-INTERFACE [NEXT-HOP]} [AD]`

There are also three different types of *static routes* that are used in both IPv4 and IPv6:

1. **Directly attached**
 - Only the *exit-interface* is specified
2. **Recursive**
 - Only the *next-hop* is specified
3. **Fully specified**
 - Both the *exit-interface* and *next-hop* are specified

IPv6 Ethernet Directly Attached routes

In IPv6, you cannot use directly attached *static routes* if the interface is an *Ethernet* interface.

- The router will let you enter the command and it'll look like it worked, but it will not be active and won't be used for routing.
- You'd have to use a **recursive** or **fully specified** route instead.

Link-Local Next-Hops

For a link-local address to be used as a *next-hop*, you have to also specify the *exit-interface* or it'll not work.

The command would be:

`ipv6 route PREFIX/PREFIX-LENGTH EXIT-INTERFACE NEXT-HOP-IP`